PACKAGE CLAUSE

You can use package clause in a file to let Go know that which package that file belongs to.

package clause ◀- - -

this should be the first code

it can only appear once

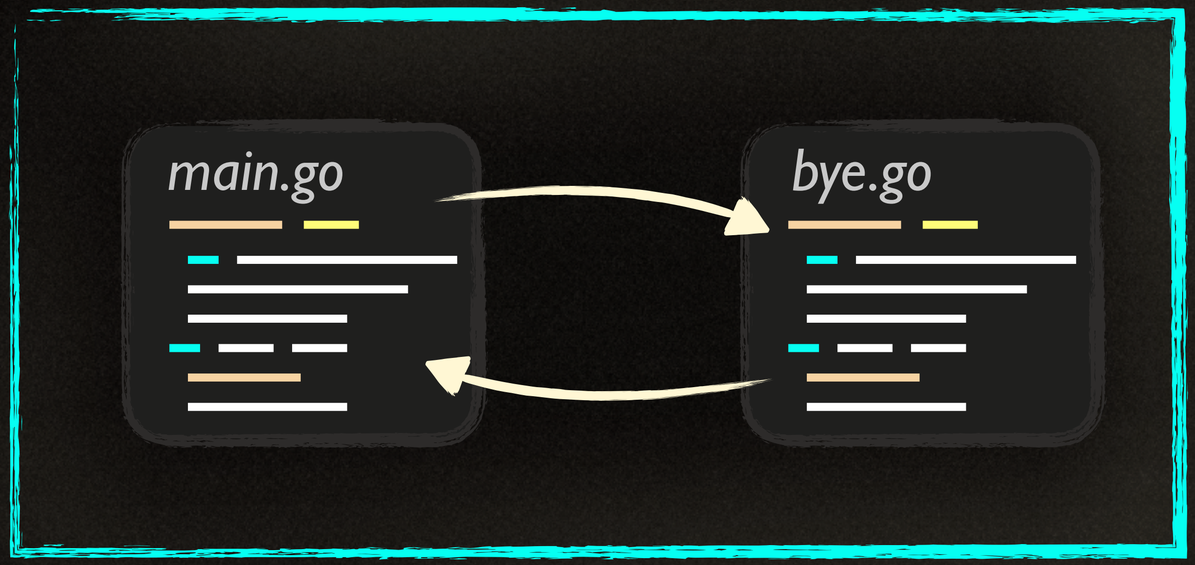
```
package main
import "fmt"

func main() {
  fmt.Println("Hello!")
}
```

PACKAGE SCOPE

Declarations which are outside of functions are visible to the files belong to the same package.

main package

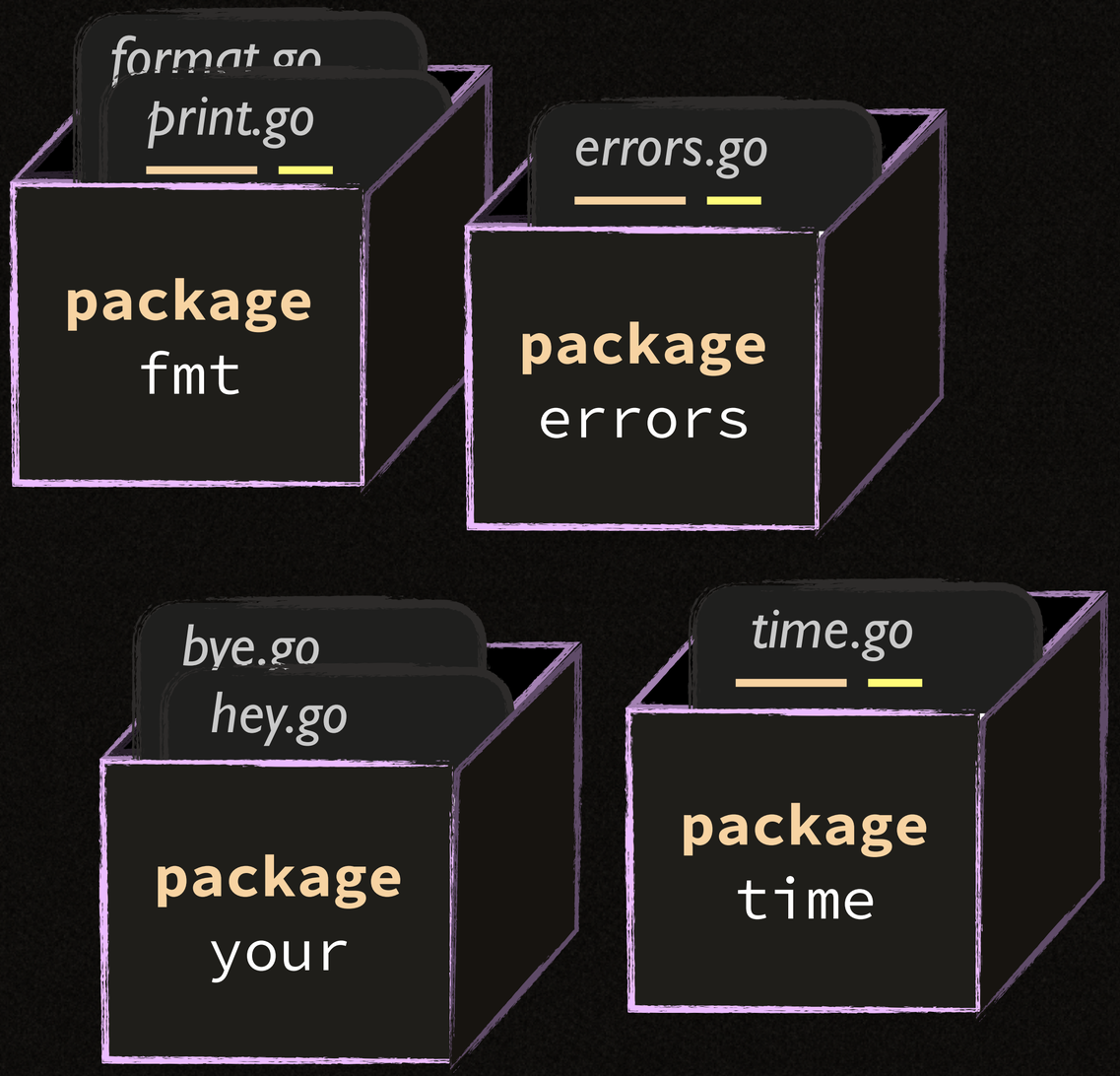


IMPORTING

Importing is like as if you've declared what's inside the imported package's files in your own file.

myfile.go

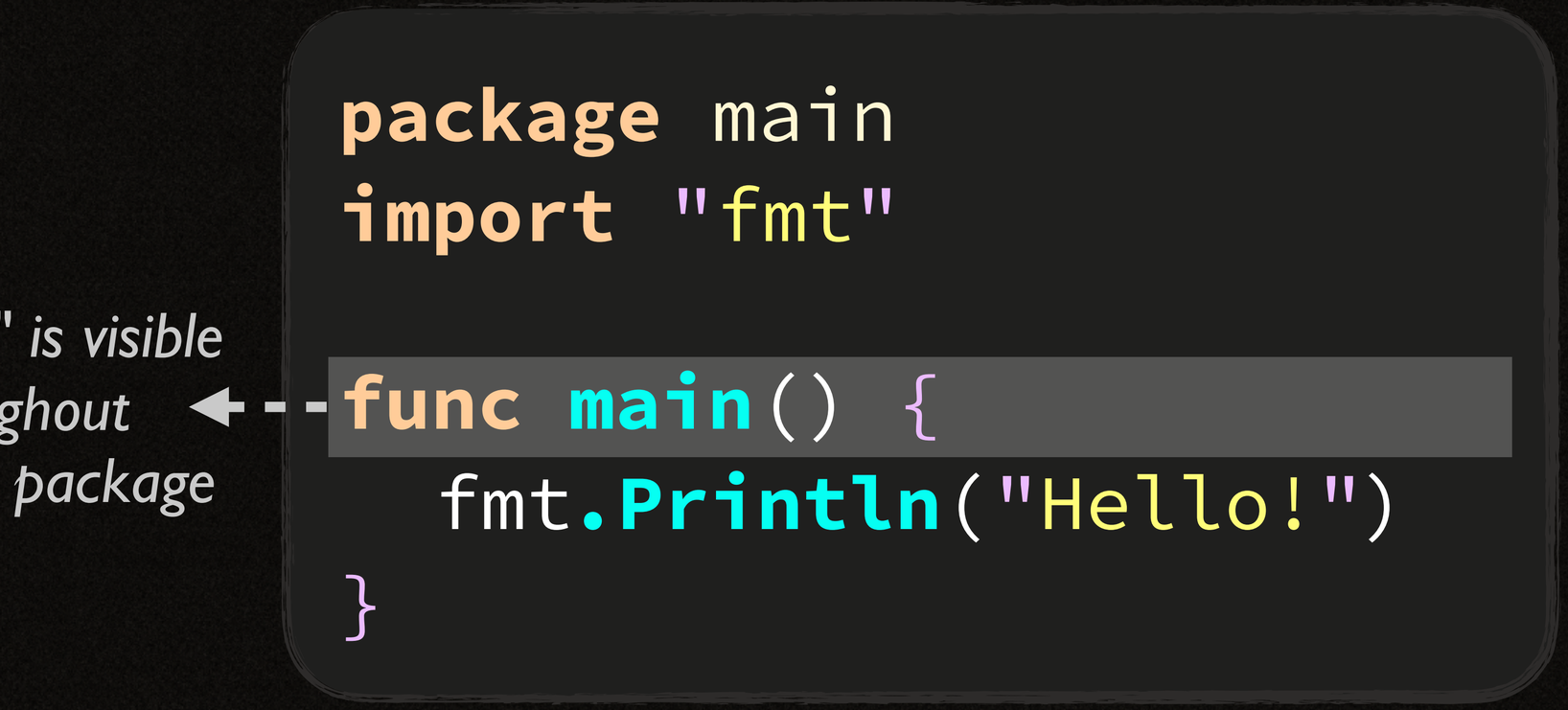
```
import "fmt"
import "errors"
import "time"
```



PACKAGE SCOPE

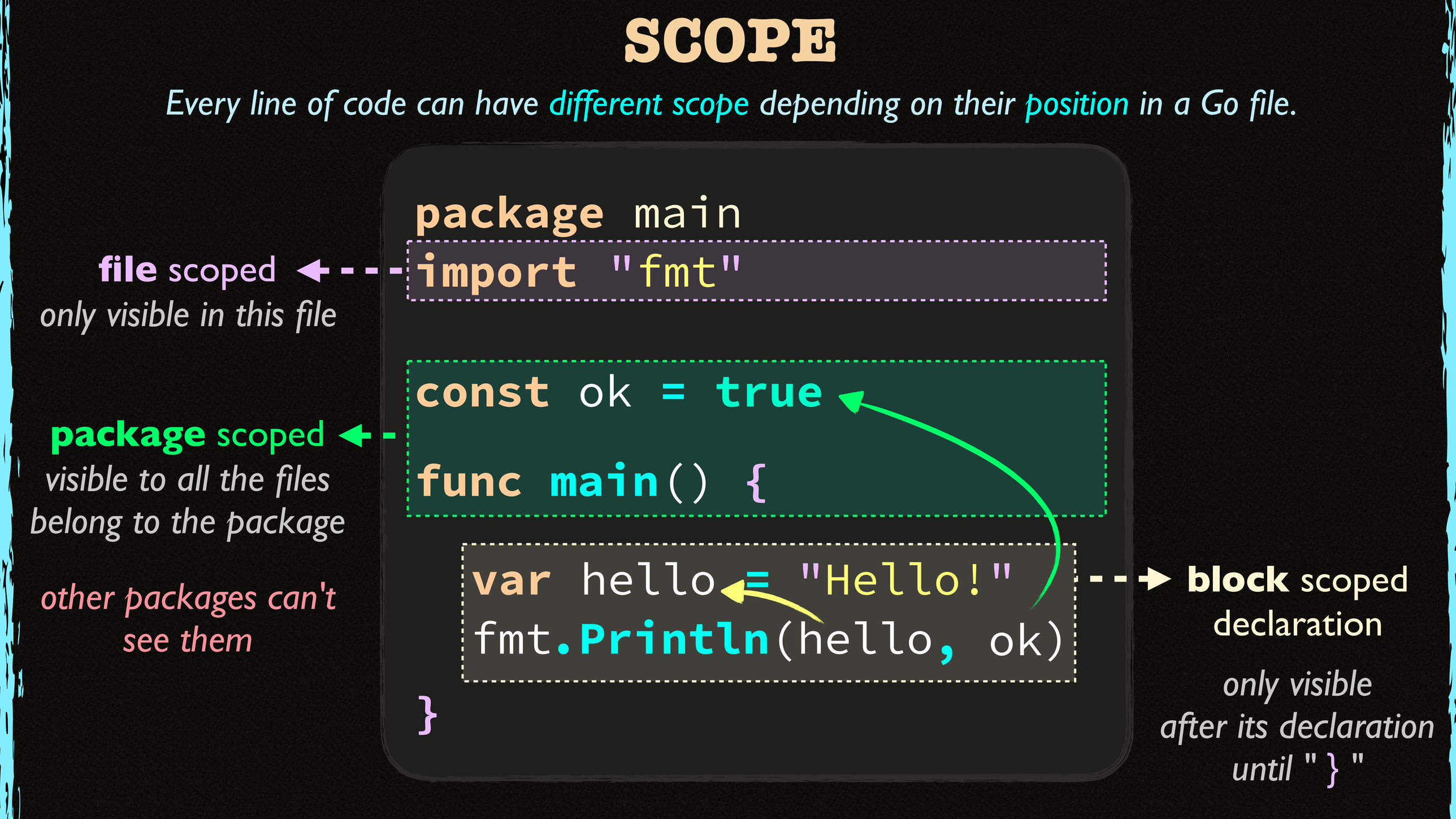
Each Go package has its own scope. For example, declared funcs are only visible to the files belong to the same package.

*"**main**()" is visible throughout the main package*



SCOPE

Every line of code can have different scope depending on their position in a Go file.



SCOPE

Every line of code can have different scope depending on their position in a Go file.

```
package main
import "fmt"

func nope() {
    const ok = true
    var hello = "Hello!"
    _ = hello
}

func main() {
    fmt.Println(hello, ok)
}
```

→ **block** scoped declaration

only visible in "nope" func

PACKAGE SCOPE

Each Go package has its own scope
For example, declared funcs are visible in the same package

main.go

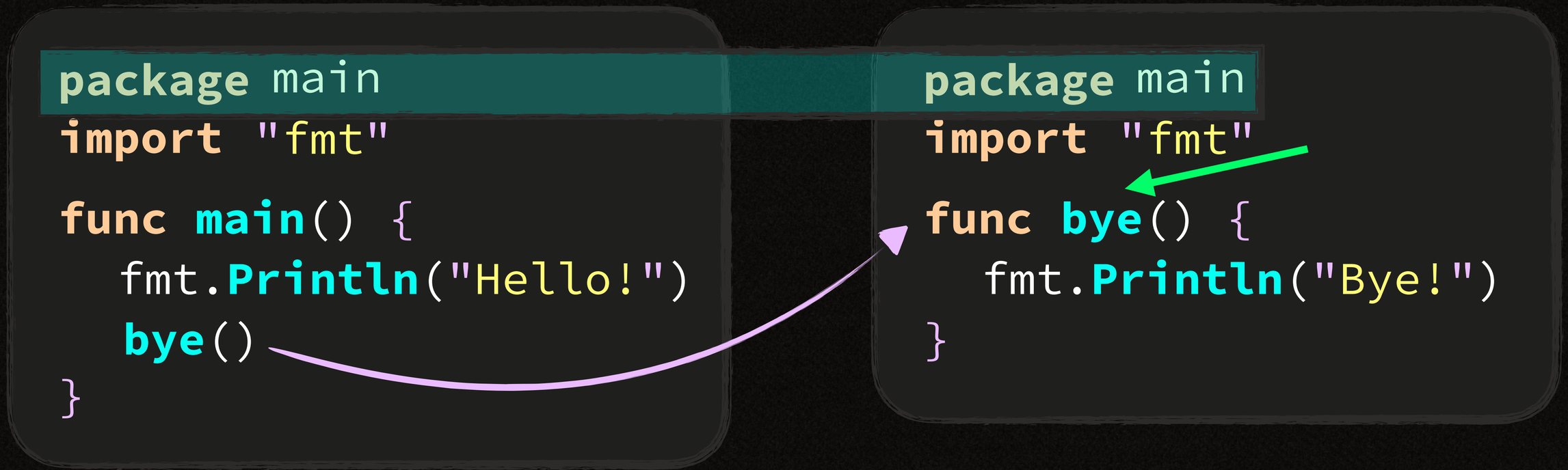
```go
package main
import "fmt"

func main() {
    fmt.Println("Hello!")
    bye()
}
```

bye.go

```go
package main
import "fmt"

func bye() {
    fmt.Println("Bye!")
}
```



FILE SCOPE

Each Go file has its own scope
Imported packages are only visible to the importing file

main.go

"***fmt***" *is visible throughout the file*

```go
package main
import "fmt"

func main() {
    fmt.Println("Hello!")
}
```

FILE SCOPE

Each Go file has its own scope
Imported packages are only visible to the importing file

bye.go

```go
package main

func bye() {
    fmt.Println("Bye!")
}
```

bye.go can't use a package that it didn't import

FILE SCOPE

Each Go file has its own scope
Imported packages are only visible to the importing file

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```
package main
import "fmt"

func bye() {
    fmt.Println("Bye!")
}
```

FILE SCOPE

Each file has to import external packages on its own

main package

learngo/first/explain/importing

PACKAGE KINDS

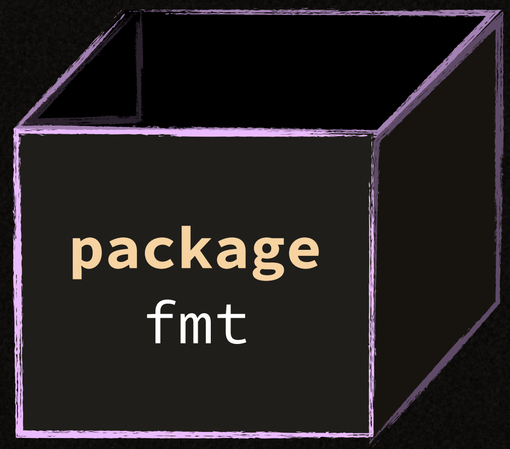
There are two kinds of packages in Go: Executable and Library.

Executable Package

+

func main( )

Library Package



LIBRARY	EXECUTABLE
created for **reusability**	created for **running**
non-executable	executable
importable	**non-importable**
can have **any name**	name should be **main**
no **func main**	**func main**